- ◇ **Title of the Project**

- **Book Recommendation System using Collaborative Filtering**

- ◇ **Objective**

- The goal of this project is to build a recommendation engine that suggests books to users based on their reading preferences and behavior. The system combines both **Popularity-Based Filtering** and **Collaborative Filtering** to enhance user experience by recommending top-rated and similar books.

- ◇ **Dataset Used**

- The dataset used is the **Book-Crossing Dataset**, which contains:

- **Books** dataset: Details about the books (Title, Author, ISBN, Image-URL)

- **Users** dataset: Basic demographic information about users

- **Ratings** dataset: User ratings for various books

- Source: Book-Crossing Dataset (from GroupLens)

- ◇ **Model Chosen**

- Two main approaches are implemented:

- **Popularity-Based Recommendation**
  Recommends top-rated books that have been rated by a large number of users.

- **Collaborative Filtering (Memory-Based - User/User Similarity)**
  Uses **Cosine Similarity** to find similar users or books based on user-book rating matrix. The model recommends books similar to a selected book using matrix-based collaborative filtering.

- Libraries used:

- Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib

- ◇ **Performance Metrics**

- As the system is not supervised, traditional accuracy metrics are not used. Instead, qualitative performance is evaluated by:

- **Relevance of Recommendations**

- **User Similarity Accuracy**

- **Diversity of Suggested Books**

- **Manual Testing on Sample Inputs**

- ---

- ◇ **Challenges & Learnings**

- **Challenges Faced:**

- Handling sparse data (many users rate very few books)

- Cleaning and merging datasets with inconsistent formats

- Ensuring the system recommends meaningful and diverse books

- **Learnings:**

- Better understanding of **Collaborative Filtering**

- Importance of data preprocessing in recommendation systems

- Working with real-world messy datasets

- Using cosine_similarity and building user-item matrices

- Creating an end-to-end ML-powered web project